



Experimental and kinetic modeling study of ammonia/dimethoxymethane oxidation in a jet-stirred reactor using SVUV-PIMS

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ABSTRACT

This study investigates the oxidation characteristics of NH_3/DMM_1 (dimethoxymethane, $\text{CH}_3\text{OCH}_2\text{OCH}_3$) blended fuels at ND-75 (1.5 % NH_3 : 0.5 % DMM_1) and ND-50 (1 % NH_3 : 1 % DMM_1 (balance O_2/Ar , $\varphi = 1$)) mixing ratios across a temperature range of 550–1100 K, employing experimental and numerical approaches to elucidate their interaction mechanisms. Experiments were conducted in a jet-stirred reactor (JSR) coupled with synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS), enabling the online detection and quantification of 17 products, including C_0 – C_3 small molecules, NH_3 , nitrogen oxides, and four rarely reported carbon-nitrogen (C–N) intermediates: HCN , HNCO , CH_3CN , and CH_3NO_2 . A detailed kinetic model, comprising 210 species and 1595 elementary reactions, was developed based on experimental data and validated against measured species profiles, demonstrating good predictive accuracy for ignition delay times and intermediate evolution. The results reveal that DMM_1 oxidation generates reactive radicals (e.g., OH , HO_2 , CH_3) that enhance NH_3 reactivity, with temperature-dependent shifts in C–N bond formation pathways—favoring $\text{CH}_3 + \text{NO}_2 = \text{CH}_3\text{NO}_2$ at 870 K and $\text{CH}_3 + \text{NH}_2 = \text{CH}_3\text{NH}_2$ at 1010 K. These intermediates play a crucial role in shaping the reaction network, providing new insights into the coupled nitrogen-carbon chemistry relevant to low-emission combustion technologies.

Novelty and significance statement

This study integrates advanced photoionization diagnostic techniques with comprehensive kinetic model development to enable full-process validation—from experimental detection to reaction mechanism elucidation. The investigation systematically examines the formation and evolution of carbon–nitrogen (C–N) intermediates within the NH_3 /dimethoxymethane (DMM_1 , $\text{CH}_3\text{OCH}_2\text{OCH}_3$) co-combustion system. Detailed analysis reveals the temperature-dependent characteristics of C–N bond formation pathways and their significant influence on NO_x evolution, addressing a critical gap in existing models that lack validation for key nitrogen-containing intermediates. This work enhances the fundamental understanding of carbon–nitrogen coupling chemistry in ammonia-fuelled combustion, offering novel insights into low-emission combustion strategies.

1. Introduction

The global drive toward carbon neutrality has intensified the search for carbon-free alternative fuels capable of replacing or supplementing conventional fossil fuels in internal combustion engines, gas turbines, and industrial furnaces. Among the various candidates, ammonia (NH_3) has emerged as a promising carbon-free fuel owing to its high hydrogen content, zero carbon dioxide (CO_2) emissions during combustion, and mature infrastructure for production, storage, and distribution [1–4]. Moreover, liquid NH_3 offers a higher volumetric energy density than hydrogen, methanol, or ethanol, making it attractive for large-scale energy deployment [5]. Nevertheless, the use of neat NH_3 is hindered by its intrinsically poor combustion characteristics, including a low laminar flame speed, high autoignition temperature, long ignition delay time, and a strong propensity to form nitrogen oxides (NO_x) [6,7].

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To address these limitations, blending NH_3 with more reactive fuels has been extensively explored as a strategy to improve combustion performance. Previous studies have investigated NH_3 mixtures with hydrogen [8–10], methane [11–13], methanol [14–18], and dimethyl ether (DME) [19–22]. These additives enhance NH_3 reactivity by enriching the radical pool—particularly OH, H, and CH_3O —thus facilitating the conversion of NH_3 to NH_2 , a critical step in its oxidation pathway. Nonetheless, several co-fuels present challenges: hydrogen is expensive and difficult to store, methanol can generate undesirable intermediates such as formaldehyde [23], and DME, despite its high cetane number and ability to accelerate NH_3 combustion, suffers from low viscosity and poor lubricity, which may compromise fuel system durability [24].

As a biomass-derived oxygenate, dimethoxymethane (DMM_1 , $\text{CH}_3\text{OCH}_2\text{OCH}_3$) possesses a high oxygen content, good volatility, and smokeless combustion characteristics, and has long been regarded as a promising low-carbon co-fuel for combustion applications [25–27]. Its high reactivity and clean-burning characteristics make it a suitable candidate for blending with NH_3 to enhance ignition and oxidation. However, compared to more extensively studied NH_3/DME or NH_3/H_2 blends, the combustion behavior of NH_3/DMM_1 mixtures remains insufficiently understood. Recent studies have begun to explore NH_3/DMM_1 combustion in limited contexts. Elbaz et al. [28] constructed a coupled kinetic mechanism for NH_3/DMM_1 and investigated premixed flame behavior based on Markstein length measurements in a spherical combustion chamber. Nonetheless, the mechanism underestimated the laminar flame speed of NH_3/DMM_1 mixtures under lean conditions. Dai et al. [29] measured ignition delay times of NH_3/DMM_1 mixtures in a shock tube at pressures of 1 and 10 bar, with an equivalence ratio of 0.5 and temperatures ranging from 1193 to 1852 K. Zhang et al. [30] developed a reaction mechanism that can predict NH_3/DMM_1 combustion characteristics, proposed a relatively complete reaction pathway, and analyzed NO_x emission behavior under different DMM_1 blending ratios. Dong et al. [31] proposed a new kinetic mechanism for NH_3/DMM_1 , calculating rate constants for key elementary reactions between DMM_1 and HNO , as well as DMM_1 and NO , and demonstrated that the mixing ratio and equivalence ratio significantly influence combustion and emission behavior. However, their model lacks experimental validation against intermediate nitrogen species.

Although previous experimental studies have provided important insights into ignition delay, flame propagation, and global emission characteristics, detailed speciation measurements under well-controlled conditions are still lacking. In particular, there remains a significant gap in validating existing kinetic models against intermediate nitrogen-containing species such as HCN , HNCO , CH_3CN , and CH_3NO_2 . Such validation is essential for accurately capturing the coupled nitrogen–carbon chemistry that governs pollutant formation and reactivity. A comprehensive understanding of these interactions is critical for the development of predictive combustion models and for enabling advanced ignition control strategies in low-emission engine technologies [32]. Moreover, many existing mechanisms rely on analogy-based estimations for key reaction rate constants rather than *ab initio* calculations or experimental measurements. Disproportionation reactions involving NH_2 and oxygenated intermediates, which can substantially affect ignition behavior and NO_x formation, are often missing or underrepresented in current models [33]. These limitations hinder the predictive accuracy of existing kinetic mechanisms, particularly in the high- to intermediate-temperature regime relevant to advanced engine technologies.

Therefore, this study focuses on the evolution of NO_x and carbon–nitrogen (C–N) intermediates during the oxidation of NH_3/DMM_1 mixtures. Oxidation experiments were conducted using a jet-stirred reactor (JSR) coupled with advanced synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) at atmospheric pressure, with an equivalence ratio of 1, over a temperature range of 550–1100 K. A detailed kinetic model for NH_3/DMM_1 combustion was

developed and evaluated through comprehensive analysis, including species identification, quantitative measurements, rate-of-production (ROP) analysis, and sensitivity analysis. The updated model shows good agreement with experimental data, accurately predicting ignition delay times (IDTs) and the evolution of key intermediates.

2. Experimental methods

The oxidation experiment of the NH_3/DMM_1 mixture was carried out at the Atomic & Molecular Physics Beamline (BL09U) of the National Synchrotron Radiation Laboratory in Hefei, China. The details of the experimental apparatus have been described in a recent publication by Xu et al. [34]. Only a brief introduction is given here. The reactor used in this work is a spherical jet stirrer reactor (JSR) made of fused silica, with an internal volume of approximately 78 cm^3 . Four counter-current nozzles are used to achieve efficient mixing of the reactants, with each nozzle having a diameter of 0.3 mm, producing a uniformly distributed jet stream. The reactants are introduced into the JSR via a quartz preheating tube with an outer diameter of 6 mm, an inner diameter of 2 mm, and a length of 75 mm. The volume of the preheating tube accounts for only 0.3 % of the JSR's volume. Due to the extremely short residence time, any reactions occurring in the preheating section can be considered negligible. The reactor is thermally controlled using a tubular furnace, with a total heating length of approximately 210 mm and a central uniform temperature zone length of approximately 150 mm. During the experiment, the temperature inside the reactor is monitored using a K-type thermocouple wrapped in molten silica to avoid catalytic effects. The measurement uncertainty is within ± 10 K. The oxidation products of the ammonia/dimethyl ether mixture were sampled through a quartz nozzle with a pore size of 130 μm welded to the sphere of the JSR, forming a molecular beam into the photoionization chamber through a skimmer, photoionized by the synchrotron radiation light and then analyzed by reflector time-of-flight mass spectrometer (RTOF-MS) [35–37]. Table 1 lists the gas composition of the reactant mixture used in the experimental procedures of this study.

In this study, oxidation experiments of the NH_3/DMM_1 mixture were conducted at atmospheric pressure. The gas residence time, defined as the ratio of reactor volume to volumetric flow rate, was 2.0 s. Species measurements were carried out over the temperature range of 550–1100 K using selected photon energies of 10.50, 11.00, 11.50, 13.20, 14.00, and 14.50 eV. The photoionization cross section (PICS) data used for the quantitative analysis of oxidation products in this work were obtained from the online database [38] and the Leiden Database [39]. Uncertainties of the experimental mole fractions were estimated within 30 % for species with known PICSs and a factor of 2 for those with estimated PICSs [40].

3. Chemical kinetics model development and validation

In this study, the Perfectly Stirred Reactor (PSR) module in CHEMKIN-PRO was employed to simulate the oxidation behavior of NH_3/DMM_1 mixtures [41]. Recently, Pan et al. [42] significantly improved the predictive accuracy of NH_3 oxidation across various equivalence ratios by updating several key rate constants in the NH_3 oxidation mechanism originally proposed by Glarborg et al. Meanwhile, the $\text{NH}_3/\text{PODE}_2$ oxidation model developed by Han et al. [43] demonstrated strong predictive performance for PODE-type oxygenates. Building on these foundations, a detailed chemical kinetic mechanism

Table 1
Experimental conditions of the NH_3/DMM_1 blends oxidation in JSR.

Fuel blend	Equivalence ratio	NH_3 (mol %)	DMM_1 (mol %)	O_2 (mol %)	Ar (mol %)
ND-50	1	1	1	4.75	93.25
ND-75	1	1.5	0.5	3.125	94.875

for NH_3/DMM_1 combustion was developed in this work, comprising 210 species and 1595 elementary reactions. The updated NH_3 oxidation framework was adopted from Pan et al. [42], and the DMM_1 sub-mechanism was integrated from Han et al. [43]. Reaction sub-mechanisms involving C_1 – C_2 hydrocarbons and NH_3 were sourced from Glarborg et al. [32], while those for CH_3CN and CH_3ONH_2 reacting with NH_3 were taken from Dai et al. [29].

Table 2

H-abstraction rate coefficients for DMM with NH_2 and NO_2 (R01–R06) were estimated by analogy to the corresponding DME reactions reported by Dai et al. [44] and applied by Han et al. [43]. Additionally, recent quantum chemical calculations by Dong et al. [31] for $\text{DMM}_1 + \text{NO}$ and $\text{DMM}_1 + \text{HNO}$ reactions (R07–R10) were incorporated into the mechanism to enhance the model's fidelity in capturing nitrogen–oxygen interactions. Rate constants (e.g., R01–R06) estimated by analogy may involve uncertainties. Future high-level quantum chemical calculations could help refine these parameters to improve model accuracy, especially in high-temperature regimes where these reactions dominate.

The thermodynamic data for nitrogen-containing species and most carbon–nitrogen intermediates used in this study were primarily sourced from the NH_3 model developed by Pan et al. [42] and relevant literature [48,49]. Thermodynamic properties for DMM_1 oxidation-related species were adopted from the work of Han et al. [43]. The complete detailed kinetic mechanism and associated thermodynamic data for NH_3/DMM_1 combustion are provided in the Supplementary Material.

4. Result and discussion

4.1. Mole fraction profiles in NH_3/DMM_1 oxidation

Fig. 1 and Fig. 2 show the experimental and simulated molar fraction distributions of several important intermediates with temperature during the oxidation of NH_3/DMM_1 under two different blending ratios. A total of 17 key components were detected and quantified, including C_0 – C_3 small-molecule species such as DMM_1 , CH_4 , H_2O , CO , CH_2O (Formaldehyde), HOCHO (Formic acid), CO_2 , and CH_3OCHO (Methyl

formate). And NH_3 and some common nitrogen oxides, such as NO , N_2O , and NO_2 , were also detected in this experiment as shown in Fig. 2(a)–(d). Additionally, it is worth noting that four C–N intermediates, including HCN (Hydrocyanic acid), HNCO (Isocyanic acid), CH_3CN (Acetonitrile) and CH_3NO_2 (Nitromethane), were identified in this experiment shown in Fig. 2(e)–(h), which have been rarely mentioned in previous studies, indicating a significant interaction process between DMM_1 and NH_3 during the oxidation. It is noteworthy that due to limitations in detection methods, the intermediates identified cover only a portion of the nitrogen balance, while N_2 and several short-lived radicals (such as CH_3NH_2 , NNH , and H_2NO) account for the remaining nitrogen fate.

The mole fraction profiles of these combustion products were quantitatively analyzed and compared with the results simulated by the present kinetic model. Generally, the current model can relatively well capture the evolution process of fuels and most intermediates. However, it is worth noting that the model predicts slightly lower mole concentrations of HCN and CH_3CN , while slightly higher mole concentrations of CH_4 in the high-temperature range compared with the experimental results, but still captures the trend of molar concentration variation with temperature well. Fig. 2 highlights distinct behaviors between the two blend ratios. ND-75, with higher NH_3 content, promotes stronger nitrogen oxidation and higher NO_x/HCN formation, whereas ND-50, enriched in DMM_1 , enhances CO and CH_2O production through DMM_1 decomposition. These contrasting trends underscore the critical role of blend composition in governing reactivity and emission characteristics. Another interesting observation needs to be pointed out that the reaction activity of NH_3 in the blends is significantly enhanced and exhibits a distinct three-stage oxidation process, which is completely different from that of DMM_1 . Previous studies by Tang et al. [53] indicated that pure NH_3 undergoes almost no oxidation under stoichiometric conditions within the temperature range of 500–1100 K, which is consistent with our model predictions (as shown in the supplementary materials Figure S2(a)). While for the NH_3/DMM_1 mixture, when the blending ratio is 50 %, NH_3 begins to enter the rapid decomposition pre-oxidation stage in the temperature range of approximately 750–850 K.

However, this highly reactive trend does not maintain continuously with the increase of temperature and after the temperature rises to 870 to 1010 K, the reaction of NH_3 gradually slows down and enters a

Table 2

Rate constants for selected reactions in the sub-mechanism of NH_3/DMM_1 blends oxidation, $k = A T^n \exp(-E_a/RT)$. Units are in cm^3 , s and cal/mol.

No.	Reactions	A	n	E_a	Reference
<i>H-Abstraction Reactions of DMM₁ by NH₂/NO/NO₂/HNO</i>					
R01	$\text{DMM}_1 + \text{NH}_2 = \text{CH}_3\text{OCH}_2\text{OCH}_2 + \text{NH}_3$	1.80×10^0	3.61	4353	[43]
R02	$\text{DMM}_1 + \text{NH}_2 = \text{CH}_3\text{OCHOCH}_3 + \text{NH}_3$	0.90×10^0	3.61	4353	[43]
R03	$\text{DMM}_1 + \text{NO}_2 = \text{CH}_3\text{OCH}_2\text{OCH}_2 + \text{HONO}$	5.80×10^1	3.50	23,755	[43]
R04	$\text{DMM}_1 + \text{NO}_2 = \text{CH}_3\text{OCH}_2\text{OCH}_2 + \text{HNO}_2$	6.50×10^2	3.00	23,176	[43]
R05	$\text{DMM}_1 + \text{NO}_2 = \text{CH}_3\text{OCHOCH}_3 + \text{HONO}$	2.90×10^1	3.50	23,755	[43]
R06	$\text{DMM}_1 + \text{NO}_2 = \text{CH}_3\text{OCHOCH}_3 + \text{HNO}_2$	3.25×10^2	3.00	23,176	[43]
R07	$\text{DMM}_1 + \text{HNO} = \text{CH}_3\text{OCHOCH}_3 + \text{H}_2\text{NO}$	2.38×10^2	0.00	242,140	[31]
R08	$\text{DMM}_1 + \text{HNO} = \text{CH}_3\text{OCH}_2\text{OCH}_2 + \text{H}_2\text{NO}$	2.38×10^2	0.00	242,140	[31]
R09	$\text{DMM}_1 + \text{NO} = \text{CH}_3\text{OCHOCH}_3 + \text{HNO}$	2.34×10^{-12}	0.00	130,070	[31]
R10	$\text{DMM}_1 + \text{NO} = \text{CH}_3\text{OCH}_2\text{OCH}_2 + \text{HNO}$	2.34×10^{-12}	0.00	130,070	[31]
<i>Secondary Reactions</i>					
R11	$\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$	2.10×10^{12}	0.00	–497	[44]
R12	$\text{NO}_2 + \text{H} = \text{NO} + \text{OH}$	2.50×10^{14}	0.00	676	[45]
R13	$\text{CH}_3 + \text{NO}_2 = \text{CH}_3\text{O} + \text{NO}$	1.10×10^{13}	0.00	0	[46]
R14	$\text{NO}_2 + \text{OH} (+\text{M}) = \text{HONO}_2 (+\text{M})$	3.00×10^{13}	0.00	0	[32]
	Low/ Troee / 0.902	2.94×10^{25} 0.40	–3.00 1×10^{-30}	0 1×10^{30}	
R15	$\text{NH}_2 + \text{NO}_2 = \text{H}_2\text{NO} + \text{NO}$	1.21×10^{15}	0.11	–1186	[32]
R16	$\text{NH}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$	2.60×10^{18}	–2.19	455	[32]
R17	$\text{CH}_3\text{NO}_2 (+\text{M}) = \text{CH}_3 + \text{NO}_2 (+\text{M})$	1.80×10^{16}	0.00	58,500	[46]
	Low/ Troee / 0.902	1.26×10^{17} 0.183	0.00 1×10^{-30}	42,000 1×10^{30}	
R18	$\text{HCNH} = \text{HCN} + \text{H}$	6.10×10^{28}	–5.69	24,271	[47]
R19	$\text{H}_2\text{CN} = \text{HCN} + \text{H}$	6.00×10^{31}	–6.46	32,110	[42]
R20	$\text{HNCO} + \text{H} = \text{NH}_2 + \text{CO}$	3.60×10^4	2.49	2345	[44]
R21	$\text{HNO} + \text{O}_2 = \text{HO}_2 + \text{NO}$	2.00×10^{13}	0.00	16,000	[47]

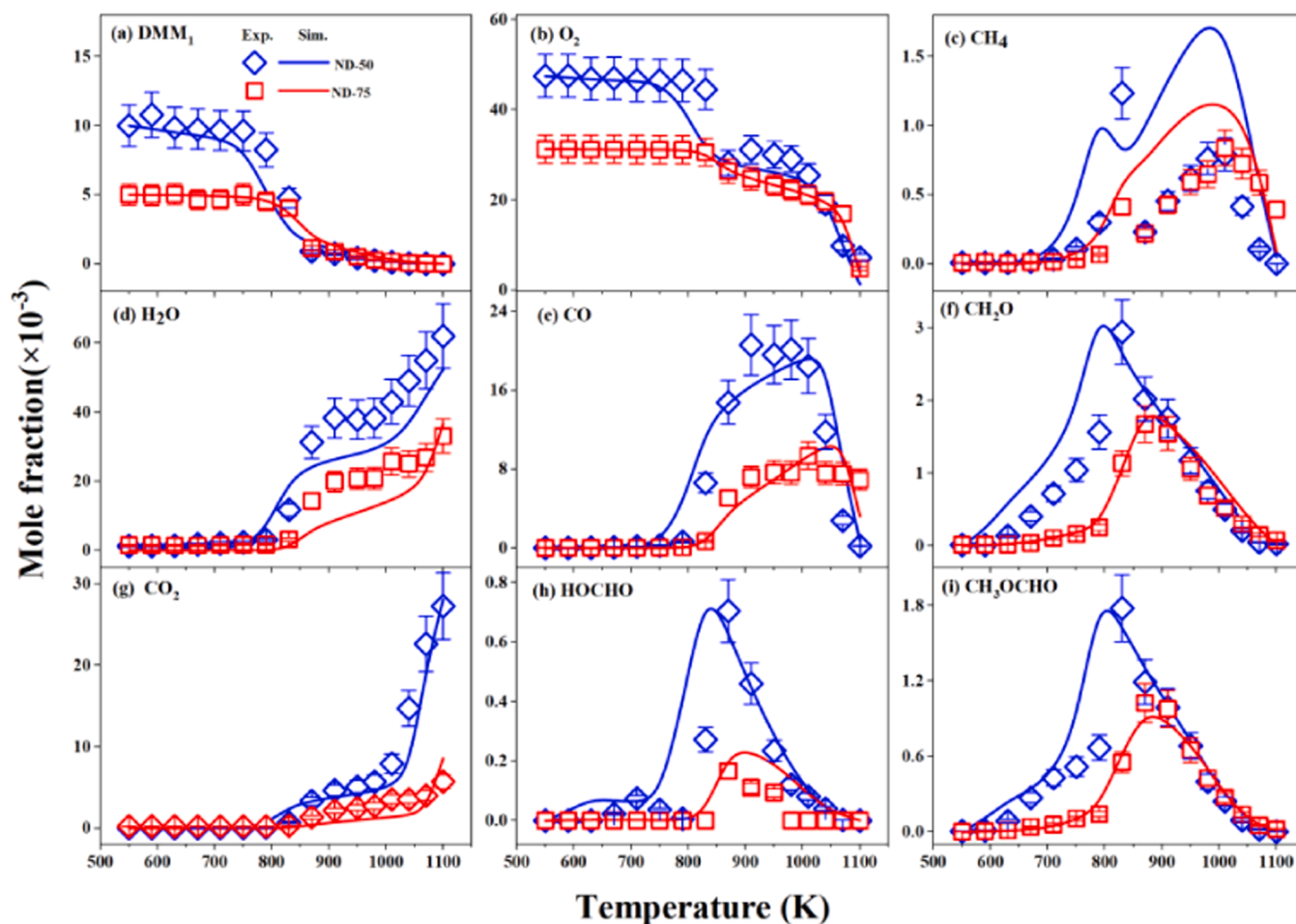


Fig. 1. Experimental and simulated molar fraction distributions of C₀–C₃ small-molecule products during the oxidation of ND-50 and ND-75. (a) DMM₁, (b) O₂, (c) CH₄, (d) H₂O, (e) CO, (f) CH₂O, (g) CO₂, (h) HOCHO, (i) CH₃OCHO. Symbols and solid lines represent experimental results and simulated results, respectively.

relative plateau zone. As the temperature further increases to 1100 K, NH₃ will enter a rapid consumption process again. The underlying mechanism of this interesting phenomenon will be discussed in detail in subsequent sections.

The present model was further validated against multiple experimental datasets, including the ignition delay times (IDTs) of NH₃/DMM₁ mixtures reported by Dai et al. [33], NH₃ IDT data from Chen et al. [51], and DMM₁ IDT data from Dias et al. [52], as shown in Fig. 3 (a–c). In general, the detailed kinetic mechanism of NH₃/DMM₁ model developed in this work can predict all the IDTs well with all three datasets, reasonably reproducing the IDTs of NH₃/DMM₁ blends, pure NH₃, and DMM₁ across a relatively wide temperature range, which reinforces the robustness of the mechanism in capturing the combustion characteristics of individual and blended fuels. In addition, the current model was validated against the experimental results of pure NH₃ oxidation and pure DMM₁ oxidation reported by Kevin et al. [54] and Tang et al. [53], respectively, to assess its accuracy, which can be obtained from Supplementary Materials Figure S1 and S2.

Overall, within the permissible range of errors, the up-to-date developed model in this study is fully and reasonably predictable for the low-to-high temperature oxidation and the IDTs of NH₃/DMM₁ blends, which can be applied to the follow-up analysis of the interaction mechanism of the blends.

4.2. Effect of DMM₁ on NH₃ reactivity

As discussed in Section 4.1, DMM₁ can significantly enhance the oxidation reactivity of NH₃ according to the JSR experimental results of

the blended fuels, and interestingly, the evolution of NH₃ exhibits three distinct stages with temperature variations. To further demonstrate the improvement effect of DMM₁ on the reactivity of NH₃/DMM₁ blends, the ignition delay times under two blend ratios (ND-50 and ND-75) and two pressures (1 atm and 40 atm) were simulated using the present kinetic model, as shown in Fig. 4. It can be seen from the results that the ignition delay time of the ND-50 mixture can be shortened by approximately 70 % compared to that of the ND-75 under the conditions of 1 atm and 1010 K. At 40 atm and 1010 K, the ignition delay time of ND-50 can even be shortened by over 80 % compared to that of ND-75. All these results indicate a potential interaction between DMM₁ and NH₃, which can significantly affect the oxidation process of NH₃ and further enhance its ignition characteristics.

To elucidate this interaction mechanism between NH₃ and DMM₁, the reaction pathway analysis and sensitivity analysis under both medium-temperature (870 K) and high-temperature conditions (1010 K) were conducted with two blend ratios of ND-50 and ND-75, as shown in Fig. 5 and Fig. 7. The numbers next to the arrows represent the percentage of each reaction pathway in the total reaction pathway flux.

The reaction pathway analysis (ROP) presented in Fig. 5 reveals that, at both temperatures of 870 K and 1010 K, NH₃ is predominantly consumed and converted into NH₂ via H-abstraction reactions with OH radicals, which are generated during the oxidation of DMM₁. Therefore, the initial oxidation temperature of NH₃ is significantly advanced, and at the blend ratio of 50 %, it even begins to oxidize and consume at only about 750 K. This can also be concluded from the sensitivity analysis of NH₃ at 870 K, as shown in Fig. 7. Due to the presence of DMM₁, the consumption of NH₃ exhibits extremely high sensitivity to the reaction

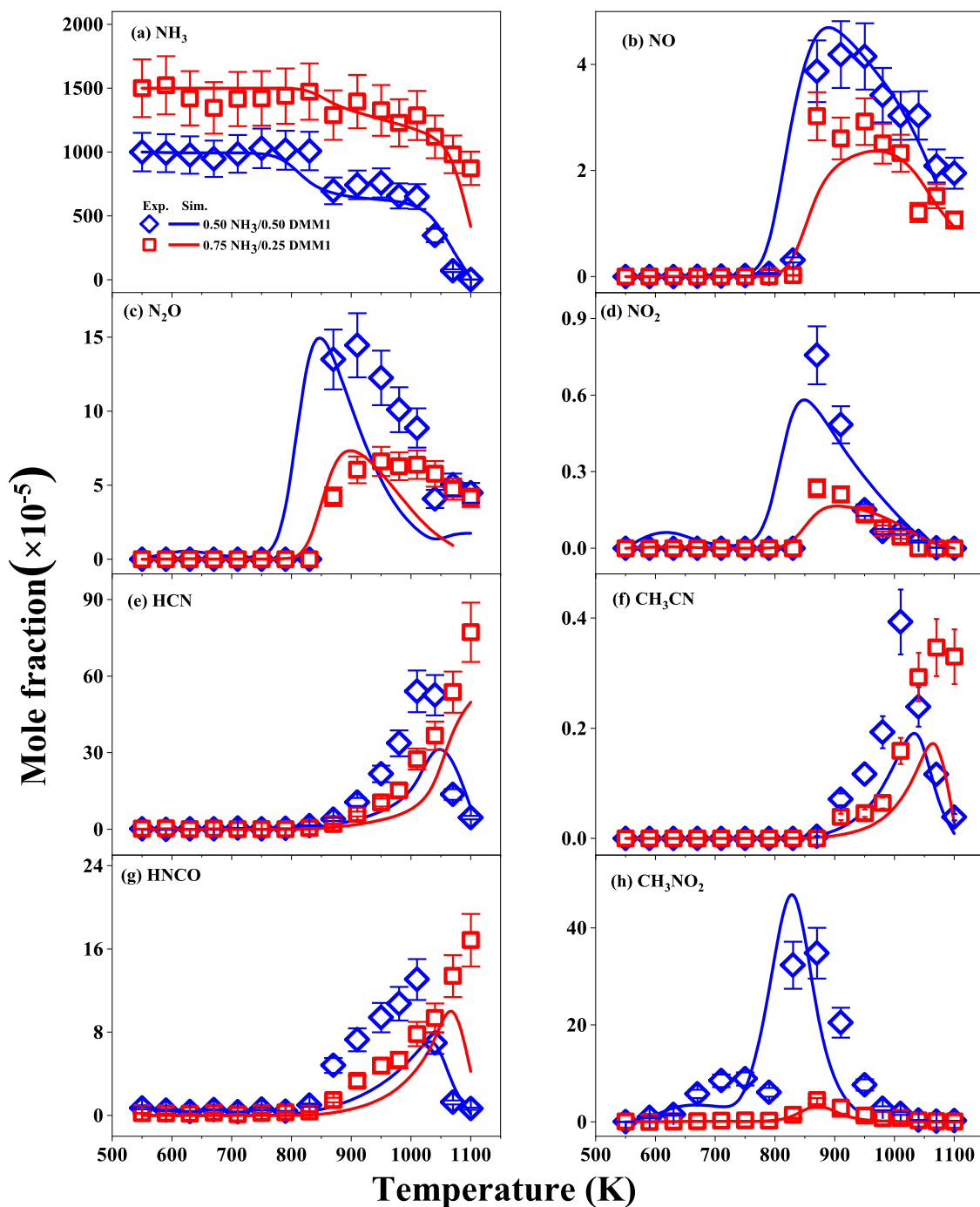


Fig. 2. Experimental and simulated molar fraction distributions of NH_3 and nitrogen intermediates during the oxidation of ND-50 and ND-75. (a) NH_3 , (b) NO , (c) N_2O , (d) NO_2 , (e) HCN , (f) CH_3CN , (g) HNCO , (h) CH_3NO_2 . Symbols and solid lines represent experimental results and simulated results, respectively.

of $\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$.

As a critical intermediate, NH_2 is mainly consumed through two pathways in the subsequent reactions, as shown in reaction pathway analysis in Fig. 5.

The majority of NH_2 will be converted to CH_3NH_2 , H_2NO , and NO through reactions with CH_3 , NO_2 , and HO_2 , respectively. However, a portion of NH_2 will still be reconverted to NH_3 via reactions with HO_2 and H_2 , i.e. $\text{NH}_2 + \text{HO}_2 = \text{NH}_3 + \text{O}_2$ and $\text{NH}_2 + \text{H}_2 = \text{NH}_3 + \text{H}$. For example, at 870 K, approximately 12 % and 21 % of NH_2 in the ND-50 and ND-75 blends, respectively, are converted back to NH_3 . Moreover, the reduction reaction between HO_2/H_2 and NH_2 gradually intensifies with increase in temperature. As illustrated in Fig. 6, within the 870–1010 K

temperature range, the concentrations of HO_2 and H_2 in the system rise with increasing temperature, leading to over 37 % of NH_2 being regenerated into NH_3 via the aforementioned pathways at 1010 K (ND-50: approximately 37.1 %; ND-75: approximately 37.9 %). This can also be confirmed from the sensitivity analysis of NH_3 , as shown in Fig. 7. At 870 K, the reaction $\text{NO}_2 + \text{H} = \text{NO} + \text{OH}$ emerges as the primary inhibitory reaction for NH_3 oxidation, while the dehydrogenation reaction of NH_3 with OH , along with the oxidation reactions of NH_2 with NO_2 and HO_2 , serve as the dominant promotional reactions. At 1010 K, the $\text{NH}_2 + \text{HO}_2 = \text{NH}_3 + \text{O}_2$ reaction, which facilitates NH_3 regeneration, becomes the leading inhibitory reaction, consistent with the ROP findings. At this elevated temperature, the oxidation of NH_2 with HO_2 exerts the

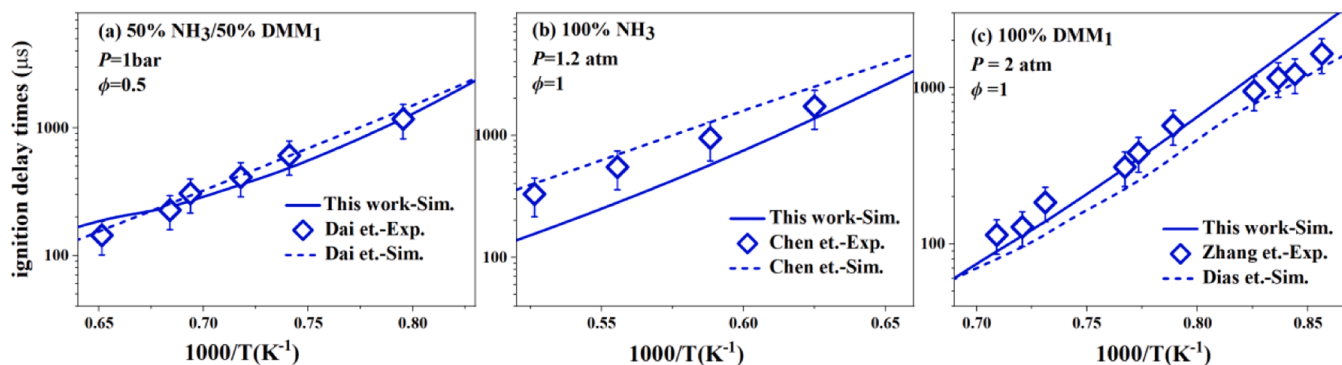


Fig. 3. Panels (a), (b), and (c) present the validation of the present model against ignition delay time (IDT) measurements for NH_3/DMM_1 blends, neat NH_3 , and DMM_1 , respectively. Solid and dashed lines represent the simulation results from the present model and from the mechanisms of Dai et al. [29], Chen et al. [50], and Zhang et al. [52], respectively. Symbols denote the corresponding experimental IDT data reported by Dai et al. [29], Chen et al. [50], and Dias et al. [51].

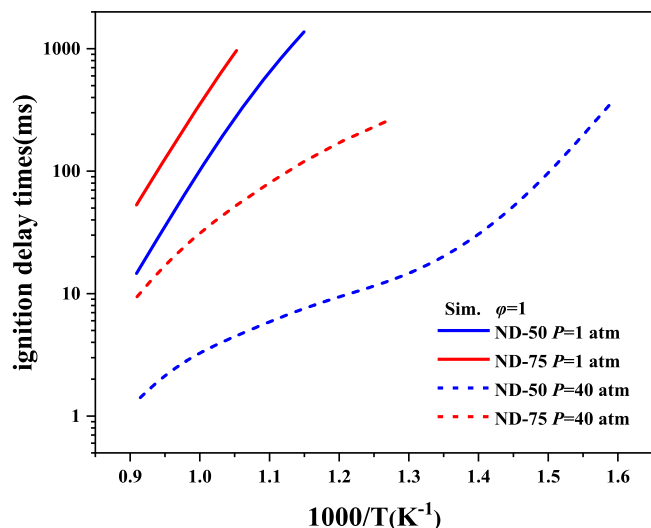


Fig. 4. Simulated ignition delay times of ND-50 and ND-75 mixtures under stoichiometric conditions at 1 atm and 40 atm pressures using the current model.

strongest promotional influence on NH_3 oxidation, complemented by the reaction $\text{H} + \text{O}_2 = \text{O} + \text{OH}$ and the dehydrogenation of NH_3 with OH , both of which enhance NH_3 oxidation. This behavior provides a mechanistic explanation for the plateau observed in the NH_3 consumption trend depicted in Fig. 2(a), underscoring the dynamic interplay between NH_3 oxidation and radical-driven regeneration processes influenced by DMM_1 blending.

4.3. Effect of DMM_1 addition on the evolution pathways of nitrogen-containing species

Another major issue of concern in the combustion process of NH_3 is the generation and evolution of NO_x . And, as is well known, fuel NO_x is one of the main sources of production during its combustion. However, in this research, analysis reveals that the addition of DMM_1 exerts a significant effect on the transformation pathways of NO_x through several C–N intermediates, which are closely related to the reactive radicals such as OH , HO_2 , and CH_3 produced by DMM_1 .

As discussed in Section 4.1, a total of four C–N intermediates, including HCN , HNCO , CH_3CN , and CH_3NO_2 , were detected in this study and exhibited significantly different mole fraction distributions under medium and high temperature conditions, indicating that the reactions related to C–N bond formation vary significantly with temperature. According to the reaction pathway (ROP) analysis in Fig. 5, under

medium temperature conditions (870 K), the formation of C–N bond is mainly formed through the addition reaction of CH_3 and NO_2 : $\text{NO}_2 + \text{CH}_3(+\text{M}) = \text{CH}_3\text{NO}_2(+\text{M})$, yielding CH_3NO_2 . While under high temperature conditions (1010 K), compared to NO_2 , CH_3 tends to react with NH_2 to form $\text{CH}_3\text{NH}:\text{NH}_2 + \text{CH}_3 = \text{CH}_3\text{NH}_2$. The CH_3 radicals in the reaction pool are primarily produced through the unimolecular dissociation reaction of $\text{DMM}_1\text{-B}$, CH_3OCH_2 , and CH_3OCO under high temperature conditions.

The primary reason for this difference is that, under moderate temperature conditions, NO_2 has stronger oxidizing properties compared to NH_2 , enabling it to more easily capture CH_3 radicals in a combustion and form CH_3NO_2 through radical recombination. Under high-temperature conditions, NH_2 is more stable than NO_2 , and as the concentration of CH_3 radicals in the system increases, NH_2 is more inclined to react with CH_3 through radical recombination to form CH_3NH_2 . Moreover, under high-temperature conditions, some NO_2 undergoes the reaction pathway $\text{NO}_2 \rightarrow \text{NO}$, converting into NO , thereby inhibiting the formation of CH_3NO_2 . Additionally, CH_3NO_2 is unstable at higher temperatures and is prone to undergo unimolecular dissociation reactions, decomposing into $\text{NO}_2 + \text{CH}_3$ again, which can be demonstrated from the mole fraction trend in Fig. 4(b) Section 4.1, where CH_3NO_2 hits its peak at 870 K and rapidly declines with subsequent increasing temperature.

C–N intermediates can significantly influence the formation and evolution of NO_x in the DMM_1/NH_3 system. ROP analysis in Fig. 8 indicates that CH_3NO_2 undergoes H-abstraction and the reaction $\text{CH}_3\text{NO}_2 + \text{H} = \text{CH}_3\text{NO} + \text{OH}$ to form CH_2NO_2 and CH_3NO , which will further produce NO through unimolecular dissociation, forming a closed-loop reaction pathway: $\text{NO} \rightarrow \text{NO}_2 \rightarrow \text{CH}_3\text{NO}_2 \rightarrow \text{CH}_2\text{NO}_2/\text{CH}_3\text{NO} \rightarrow \text{NO}$, ultimately influencing the evolution of NO .

Regarding the CH_3NH_2 , as depicted in Fig. 9, it undergoes two dehydrogenation reactions to form CH_2NH . Subsequently, CH_2NH is converted into C–N intermediates: HCNH and H_2CN through hydrogen abstraction reactions with radicals of H , OH , and NH_2 . Then, HCNH and over 50 % of H_2CN will be consumed and produce HCN through unimolecular dissociation reactions of $\text{HCNH} = \text{HCN} + \text{H}$ and $\text{H}_2\text{CN} = \text{HCN} + \text{H}$. Most HCN reacts with CH_3 : $\text{CH}_3 + \text{HCN} = \text{CH}_3\text{CN} + \text{H}$, forming CH_3CN . Subsequently, CH_3CN undergoes H-abstraction reactions and oxidation pathways: $\text{CH}_3\text{CN} \rightarrow \text{CH}_2\text{CN} \rightarrow \text{NO}$, converting to NO ultimately, which is a notable source of NO in the DMM_1/NH_3 system.

The remaining small amount of HCN will produce HNCO through the intermediate-temperature pathway (870 K): $\text{HCN} \rightarrow \text{HNC} \rightarrow \text{HNCO}$ and the high-temperature pathway (1010 K): $\text{HCN} \rightarrow \text{NCO} \rightarrow \text{HNCO}$. After the hydrogen abstraction reactions, HNCO will be completely converted into NCO . The vast majority of NCO will react with CH_2O through $\text{CH}_2\text{O} + \text{NCO} = \text{HNO} + \text{HCO}$ to generate HNCO and HCO , but there is still a small amount of NCO that will react with NO and NO_2 to generate N_2O . As one of the important NO_x pollutants, the pathway for N_2O formation in this DMM_1/NH_3 system should be taken seriously. Interestingly, the

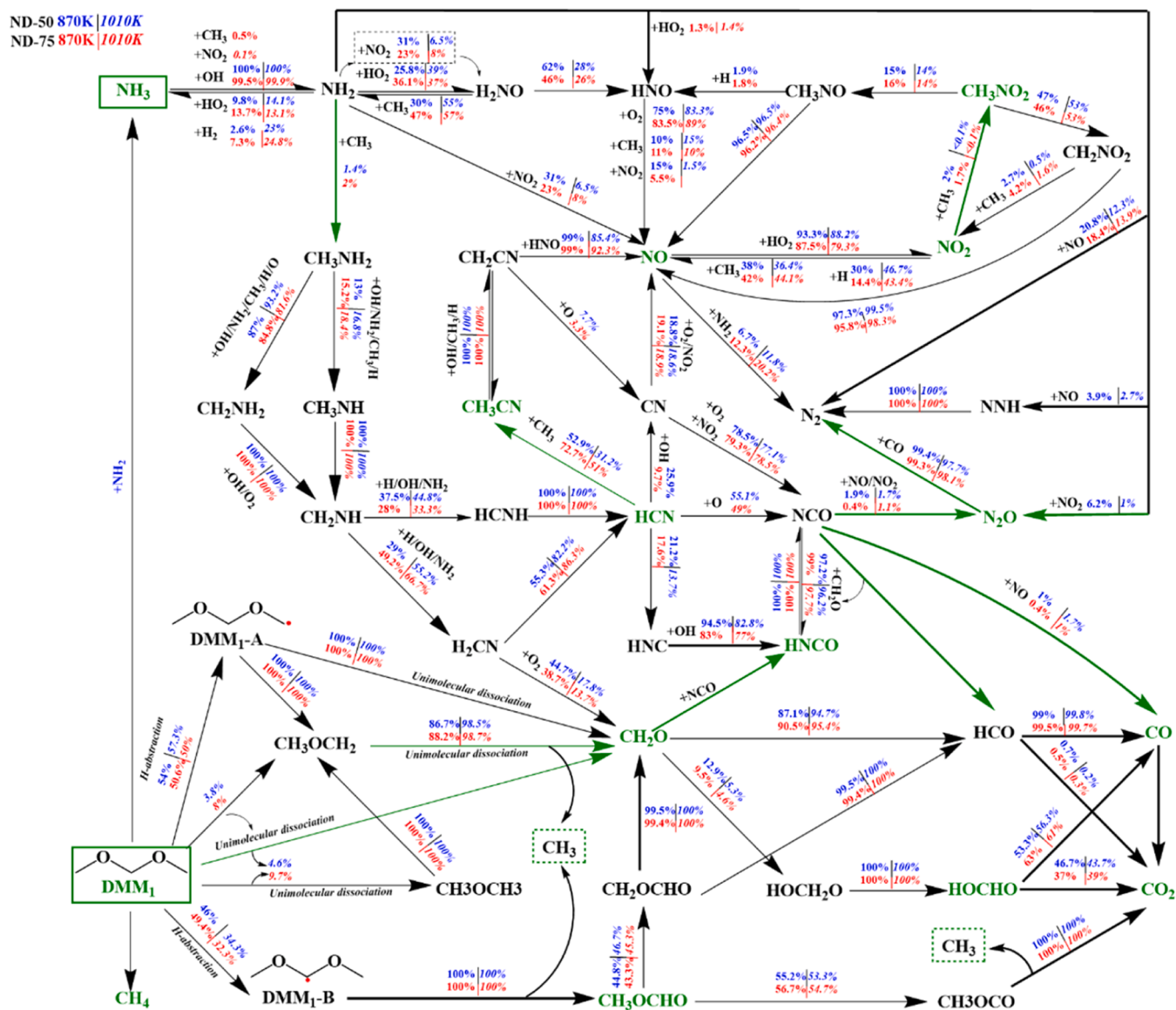


Fig. 5. Reaction path analysis of ND-50 fuel (blue), and ND-75 fuel (red) at 870 K (normal) and 1010 K (italic).

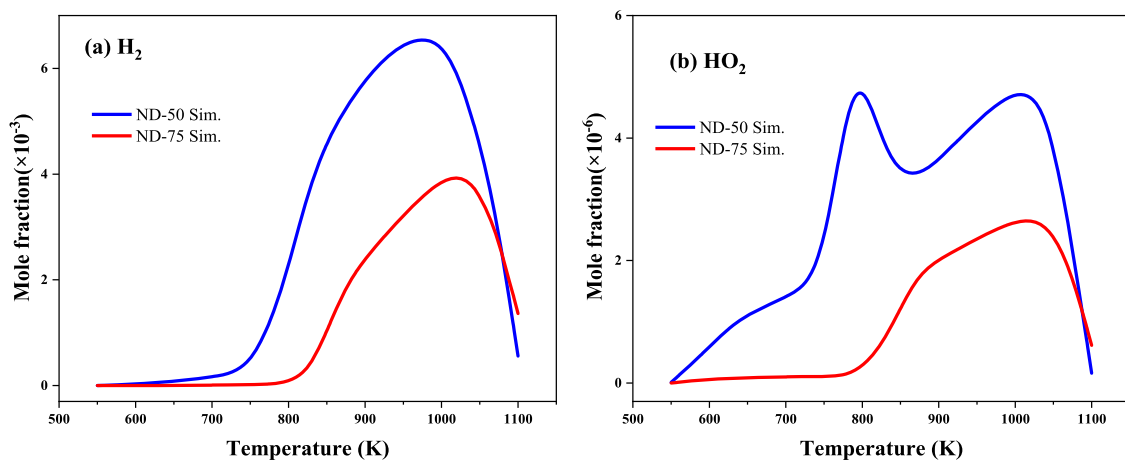


Fig. 6. Simulated mole fraction distributions of H_2 and HO_2 during the oxidation process under ND-50 and ND-75 conditions using the present model.

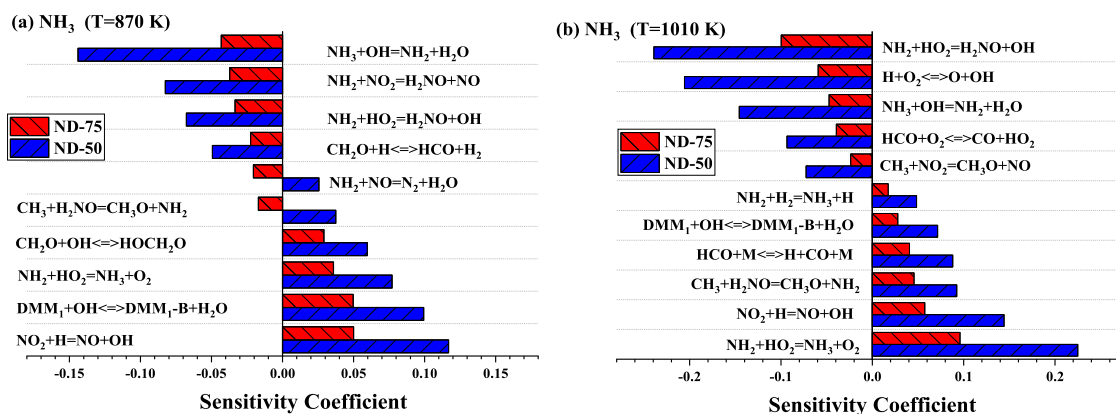


Fig. 7. Sensitivity analysis of NH_3 with blending ratios of ND-50 (blue) and ND-75 (red) was performed under conditions of $T = 870\text{ K}$ (a) and $T = 1010\text{ K}$ (b). The positive value represents inhibiting effect, and the negative value represents promoting effect.

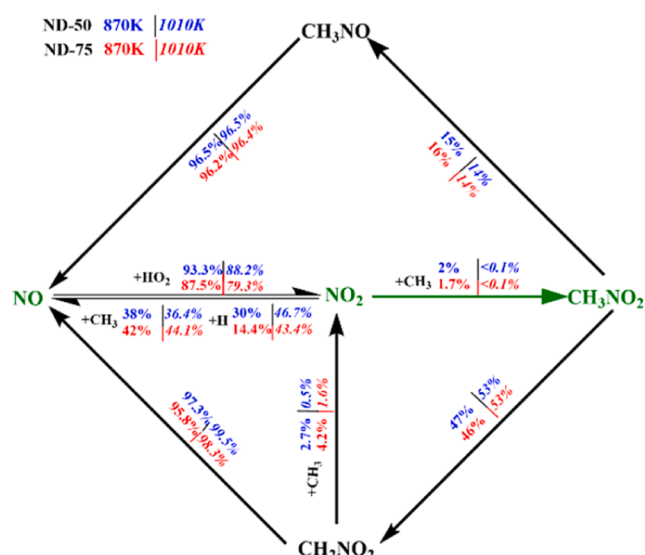


Fig. 8. Reaction pathway analysis of CH_3NO_2 at 870 K (normal) and 1010 K (italics).

analysis in Fig. 9 reveals that CO generated from DMM_1 contributes to the reduction of N_2O via the reaction $\text{N}_2\text{O} + \text{CO} = \text{N}_2 + \text{CO}_2$.

In summary, due to the unique C—O structure of DMM_1 , the large amount of CH_3 radicals and CH_2O generated during oxidation significantly promotes the interaction between carbon and nitrogen, leading to the formation of HCN, HNCO, CH_3CN , and CH_3NO_2 during the combustion of NH_3/DMM_1 blends. And, these C—N intermediates will ultimately influence the oxidation and evolution pathways of NO, N_2O , and NO_2 .

5. Conclusion

This study examines the oxidation process of NH_3/DMM_1 blended fuel at ND-75 and ND-50 ratios within the temperature range of 550–1100 K, employing both experimental and numerical simulations to elucidate its interaction mechanism. The experiments were conducted in a jet stirred reactor (JSR) and combined with synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) for online detection of the products. A total of 17 products were identified and quantified, including C₀–C₃ small molecules, NH_3 , and some common nitrogen oxides, as well as four C—N intermediates, including HCN (hydrocyanic acid), HNCO (isocyanic acid), CH_3CN (acetonitrile), and CH_3NO_2 (nitromethane). These four C—N intermediates have rarely

been mentioned in previous experiments. Based on the experimental results, a kinetic model for the oxidation of NH_3/DMM_1 mixed fuels was constructed, including 210 species and 1595 reactions. This model exhibits good consistency with experimental oxidation data. The main conclusions are as follows:

1. The addition of DMM_1 generates a substantial number of active free radicals (e.g., OH, HO_2 , and CH_3), which significantly enhance the reactivity and ignition performance of NH_3 . Especially when the blending ratio is ND-50, the oxidation temperature of NH_3 can be advanced to 750 K.
2. The mechanism by which DMM_1 influences NH_3 combustion can be divided into three temperature stages: the initial combustion stage and the high-temperature combustion stage, where NH_3 oxidation is primarily promoted through the reactions $\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$ and $\text{NH}_2 + \text{HO}_2 = \text{H}_2\text{NO} + \text{OH}$. In the intermediate-to-high temperature range of 870–1010 K, the pathway for regenerating NH_3 via the reaction $\text{NH}_3 + \text{O}_2 = \text{NH}_2 + \text{HO}_2$ is enhanced, thereby partially inhibiting NH_3 oxidation.
3. DMM_1 exhibits temperature-dependent differences in the formation of carbon-nitrogen bonds during NH_3 combustion. At 870 K, carbon-nitrogen bond formation primarily occurs via the reaction $\text{CH}_3 + \text{NO}_2 = \text{CH}_3\text{NO}_2 (+\text{M})$, while at 1010 K, the formation pathway shifts to $\text{CH}_3 + \text{NH}_2 = \text{CH}_3\text{NH}_2$. Furthermore, the evolution of these C—N intermediates (such as HCN, HNCO, CH_3CN , and CH_3NO_2) can also significantly influence the formation and transformation pathways of nitrogen oxides and related species in the system, including: $\text{NO} \rightarrow \text{NO}_2 \rightarrow \text{CH}_3\text{NO}_2 \rightarrow \text{CH}_2\text{NO}_2/\text{CH}_3\text{NO} \rightarrow \text{NO}$ and $\text{NH}_2 \rightarrow \text{CH}_3\text{NH}_2 \rightarrow \text{HCN} \rightarrow \text{HNCO}/\text{CH}_3\text{CN} \rightarrow \text{NO}/\text{N}_2\text{O}$.

Specifically, our results show that DMM_1 addition modifies nitrogen transformation pathways by temporarily storing fuel nitrogen in intermediates such as HCN and CH_3NO_2 , delaying peak NO formation. Additionally, the $\text{N}_2\text{O} + \text{CO} = \text{N}_2 + \text{CO}_2$ pathway becomes more prominent with higher DMM_1 , suggesting a potential mitigation route for N_2O . These findings underscore that selecting appropriate blend ratios and operating conditions can help balance reactivity with pollutant reduction, supporting cleaner combustion technologies.

Supplementary material

Supplementary Material is submitted along with the manuscript.

CRediT authorship contribution statement

Tenglong Lv: Writing – original draft, Investigation, Data curation. Yifeng Wang: Software, Investigation. Xin Zhong: Writing – review &

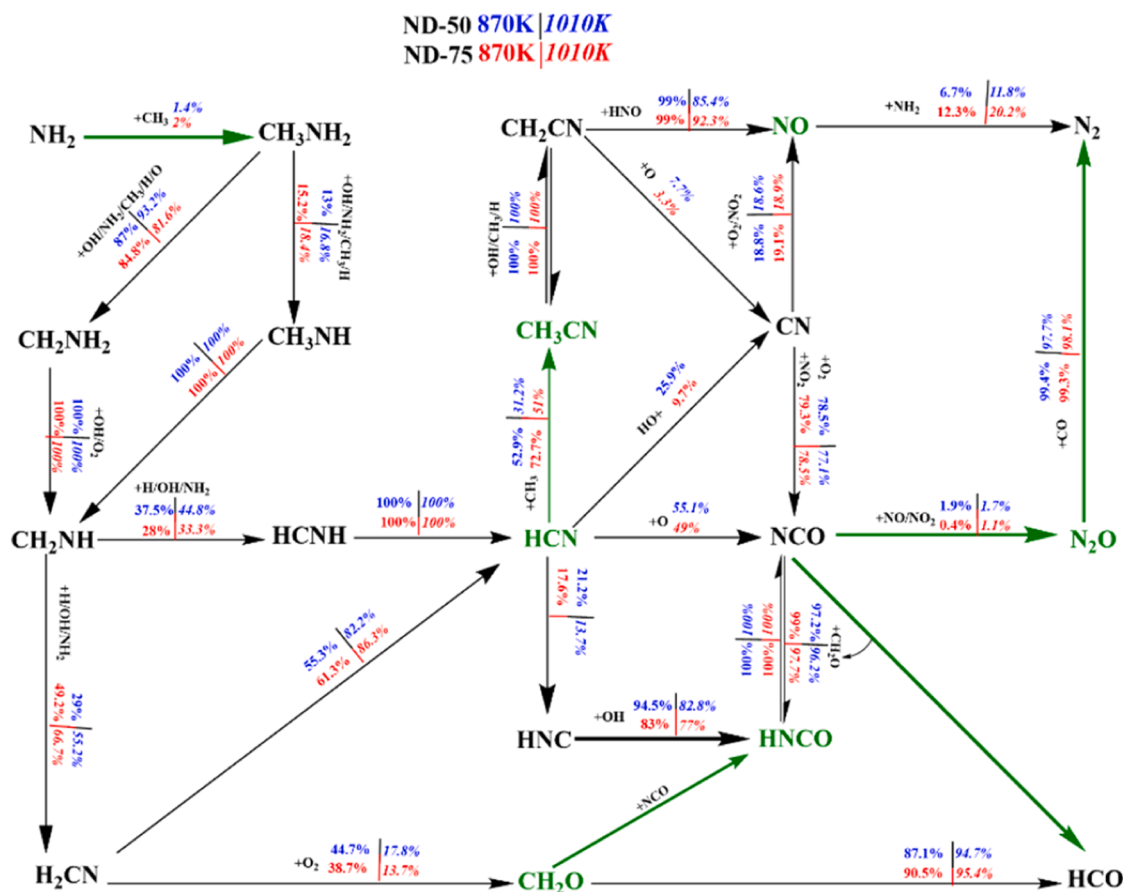


Fig. 9. Reaction pathway analysis of CH_3NH_2 at 870 K (normal) and 1010 K (italics).

editing. **Qiang Xu:** Investigation. **Jinyang Zhang:** Investigation. **Zhandong Wang:** Investigation. **Zunqing Zheng:** Supervision, Funding acquisition. **Hu Wang:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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